

Heparin (unfractionated): Drug information - UpToDate

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For additional information see "Heparin (unfractionated): Patient drug information" and "Heparin (unfractionated): Pediatric drug information"

For abbreviations, symbols, and age group definitions show table

Brand Names: Canada

- Heparin Leo

Pharmacologic Category

- Anticoagulant

Dosing: Adult

Dosage guidance:

Safety: Many concentrations of heparin are available ranging from 1 to 20,000 units/mL. Carefully examine each prefilled syringe or vial to ensure the correct concentration is selected.

Dosing: For weight-based IV heparin, an institution-specific dosing nomogram may help to achieve therapeutic anticoagulation more rapidly (see example based on aPTT under "Venous thromboembolism treatment"). If unusually large doses of heparin are required to achieve therapeutic targets, consider possible heparin resistance (Ref).

Antibiotic lock technique, adjunctive therapy

Antibiotic lock technique, adjunctive therapy (catheter-salvage strategy) (off-label use):

Note: Antibiotic lock therapy is used in addition to systemic antibiotics for certain catheter-related infections when the catheter cannot be removed. Heparin is incompatible with ethanol and should not be used with ethanol lock therapy (Ref). Heparin is also incompatible with certain antibiotics; confirm compatibility prior to use (Ref).

Intracatheter: 100 to 5,000 units/mL in combination with an appropriate antibiotic. Heparin concentration depends on compatibility with the selected antibiotic, antibiotic concentration, and catheter type, which may vary by institution (Ref). For patients with end-stage renal disease requiring hemodialysis, maximum final heparin concentration should not exceed 1,000 units/mL due to increased risk of bleeding (Ref). Instill into each lumen of the catheter access port using a sufficient volume to fill the catheter (eg, 2 to 5 mL) with a maximum dwell time of ≤ 72 hours, depending on frequency of catheter use. Withdraw lock solution prior to catheter use; replace with fresh lock solution after catheter use (Ref).

Atrial fibrillation

Atrial fibrillation (to prevent stroke and systemic embolism): Note: When admitted for short-term hospitalization (eg, admission for a procedure or surgery), ambulatory patients taking an oral anticoagulant and *not* at high risk of immediate thromboembolism typically do not require bridging anticoagulation. Patients at high risk of thromboembolism (eg, recent cardioversion, high CHA₂DS₂-VASc score, prior cardioembolic stroke, current intracardiac thrombus) may be considered for bridging with a parenteral anticoagulant (see **Transitioning between anticoagulants** below) (Ref).

IV: Initial bolus of 60 to 80 units/kg (maximum: 5,000 units), followed by a continuous infusion of 12 to 18 units/kg/hour (maximum: 1,000 units/hour). Institutional dosing protocols may vary; adjust infusion rate to maintain anticoagulation target (Ref).

Burn inhalation injury

Burn inhalation injury (off-label use): Note: Use in combination with inhaled N-acetylcysteine. Initiate within the first 48 hours of injury (Ref).

Inhalation, nebulization (off-label route): 5,000 to 10,000 units (in 3 mL) every 4 hours for 7 to 14 days or until extubation (Ref).

Extracorporeal membrane oxygenation, anticoagulation of circuit

Extracorporeal membrane oxygenation (ECMO), anticoagulation of circuit:

Note: Standard dosing has not been established for venovenous or venoarterial ECMO. Recommendations provided below are example regimens. Refer to institutional protocols. May need to individualize dose based on patient-specific needs.

IV: Initial: Bolus 50 to 100 units/kg **or** 2,500 to 5,000 units once at ECMO cannulation, followed by a continuous infusion of 8 to 12 units/kg/hour (maximum: 1,600 units/hour) (Ref). Titrate to target anti-Xa of 0.3 to 0.5 unit/mL. Alternatively, may titrate to aPTT of 50 to 80 seconds (Ref).

Frostbite

Frostbite (adjunctive agent) (off-label use): Note: Patients should be transferred to or managed in conjunction with a facility that is familiar with frostbite management. Anticoagulation may be used following administration of thrombolytic therapy (Ref). Optimal anticoagulation regimens after thrombolytic therapy have not been identified; refer to institutional protocols.

Example regimens include:

When thrombolytic is administered IV:

IV: 500 to 1,000 units/hour as a fixed dose or targeted to an aPTT 2× control for up to 6 hours (Ref); some centers extend the duration to up to 5 days (Ref) **or** transition to therapeutic low-molecular-weight heparin (LMWH) (eg, enoxaparin) (Ref).

When thrombolytic is administered intra-arterial:

Intra-arterial: 500 units/hour administered through the intra-arterial catheter; targeted to a goal aPTT 50 to 70 seconds; after discontinuation of thrombolytic therapy, continue anticoagulation for 72 to 96 hours **or** transition to therapeutic LMWH (eg, enoxaparin) (Ref). Instead of intra-arterial heparin, some suggest the use of IV heparin after intra-arterial thrombolytic (see IV dosing above) (Ref).

Hemodialysis, anticoagulation of circuit

Hemodialysis, anticoagulation of circuit:

Note: Standard dosing has not been established for intermittent hemodialysis or CRRT. Recommendations provided below are example regimens. Refer to institutional protocols. May need to individualize dose based on patient-specific needs.

Intermittent hemodialysis:

Standard risk for bleeding:

IV: Initial: Bolus 1,000 units or 2,000 units at the beginning of hemodialysis, followed by a continuous infusion of 500 units/hour; stop the infusion 60 minutes before the end of hemodialysis (Ref).

If clotting occurs during first half of hemodialysis session: Increase IV bolus by 500 units before each subsequent session (eg, administer 1,500 units or 2,500 units, depending on the previously attempted dose, at the beginning of hemodialysis) until clotting no longer occurs or up to a maximum of 4,000 units; administer continuous infusion as described above (Ref).

If clotting occurs during second half of hemodialysis session: Administer the same IV bolus dose, but increase continuous infusion by 100 units/hour for each subsequent session (eg, administer 600 units/hour) until clotting no longer occurs or up to a maximum of 1,000 units/hour; stop the infusion 30 minutes before the end of hemodialysis (Ref).

High risk for bleeding:

Note: There is no standard practice to prevent clotting of the hemodialysis circuit among patients who are at high risk for bleeding. Dialysis *without* heparin is preferred, but may consider heparin anticoagulation in patients who continue to have recurrent filter thrombosis (Ref).

IV: Initial: Bolus 1,000 units at the beginning of hemodialysis; do not administer a continuous infusion during the procedure (Ref).

If clotting occurs during hemodialysis: IV: Bolus 1,000 units at the beginning of the subsequent hemodialysis session, followed by a continuous infusion of 500 units/hour; stop the infusion 60 minutes before the end of hemodialysis; if clotting continues, increase continuous infusion by 100 units/hour at each subsequent session (eg, administer 600 units/hour) until clotting no longer occurs or up to a maximum of 1,000 units/hour (Ref).

CRRT:

Note: For patients who require anticoagulation due to repeated clotting of hemofilter and when regional citrate anticoagulation is not feasible or unavailable (Ref).

IV: Initial: Bolus 500 to 1,000 units, followed by a continuous infusion of 500 units/hour; titrate to maintain an aPTT ~1.5 times the control or ~45 seconds (Ref).

Ischemic heart disease

Ischemic heart disease:

Acute coronary syndromes:

ST-elevation myocardial infarction (off-label use):

Adjunct to percutaneous coronary intervention: see **Percutaneous coronary intervention** for dosing guidance.

Adjunct to fibrinolysis: **IV:** Bolus 60 units/kg (maximum: 4,000 units), followed by 12 units/kg/hour (maximum: 1,000 units/hour); adjust infusion rate to maintain anticoagulation target based on institutional protocol; continue for ≥48 hours, the duration of hospital stay (maximum of 8 days), or until revascularization (if performed) (Ref).

No planned reperfusion: **IV:** Bolus 50 to 70 units/kg (maximum: 5,000 units), followed by 12 units/kg/hour; adjust infusion rate to maintain anticoagulation target based on institutional protocol; continue for ≥48 hours (Ref).

Non-ST-elevation acute coronary syndromes (off-label use):

Noninvasive, ischemia-guided approach: **IV:** Bolus 60 units/kg (maximum: 5,000 units), followed by 12 units/kg/hour (maximum: 1,000 units/hour); adjust infusion rate to maintain anticoagulation target based on institutional protocol; continue for ≥48 hours, until hospital discharge, or until management changes to an invasive strategy (eg, percutaneous coronary intervention [PCI]) (Ref).

Invasive approach (adjunct to percutaneous coronary intervention): see **Percutaneous coronary intervention** for dosing guidance.

Percutaneous coronary intervention (off-label use):

No prior anticoagulant therapy:

No planned glycoprotein IIb/IIIa inhibitor use: IV: Initial bolus of 70 to 100 units/kg (maximum: 10,000 units) to achieve activated clotting time (ACT) of 250 to 300 seconds (goal ACT may vary depending on point-of-care device); repeat bolus as needed to maintain goal ACT throughout procedure (Ref).

Planned glycoprotein IIb/IIIa inhibitor use: IV: Initial bolus of 50 to 70 units/kg (maximum: 7,000 units) to achieve ACT of 200 to 250 seconds (regardless of point-of-care device); repeat bolus as needed to maintain goal ACT throughout procedure (Ref).

Prior anticoagulant therapy:

Prior anticoagulation with **heparin**:

No planned glycoprotein IIb/IIIa inhibitor use: IV: Check ACT prior to PCI and administer heparin bolus as needed (eg, 2,000 to 5,000 units) to achieve ACT of 250 to 300 seconds (goal ACT may vary depending on point-of-care device); repeat bolus (maximum: 10,000 units) as needed to maintain goal ACT throughout procedure (Ref).

Planned glycoprotein IIb/IIIa inhibitor use: IV: Check ACT prior to PCI and administer heparin bolus as needed (eg, 2,000 to 5,000 units) to achieve ACT of 200 to 250 seconds (regardless of point-of-care device); repeat bolus (maximum 7,000 units) as needed to maintain goal ACT throughout procedure (Ref).

Prior anticoagulation with **enoxaparin**:

If percutaneous coronary intervention occurs ≤12 hours after the last SUBQ dose of enoxaparin: Transition to unfractionated heparin may not be preferred to reduce the risk of bleeding complications (Ref). Refer to Enoxaparin monograph for dosing recommendations.

If percutaneous coronary intervention occurs >12 hours after the last SUBQ dose of enoxaparin: May use unfractionated heparin; refer to recommendations above for PCI with no prior anticoagulant therapy (Ref).

Prior anticoagulation with **fondaparinux**:

No planned glycoprotein IIb/IIIa inhibitor use: IV: Initial bolus of 85 units/kg (maximum: 10,000 units) to achieve ACT of 250 to 300 seconds (goal ACT may vary depending on point-of-care device); repeat bolus as needed to maintain goal ACT throughout procedure (Ref).

Planned glycoprotein IIb/IIIa inhibitor use: IV: Initial bolus of 60 units/kg (maximum: 7,000 units) to achieve ACT of 200 to 250 seconds (regardless of point-of-care device); repeat bolus as needed to maintain goal ACT throughout procedure (Ref).

Mechanical heart valve, bridging anticoagulation

Mechanical heart valve, bridging anticoagulation (for interruptions in warfarin therapy) (off-label use): Note: Bridging during intervals of subtherapeutic anticoagulation should be considered for patients with mechanical mitral, tricuspid, or pulmonary valve replacement; however, for patients with mechanical aortic valve replacement, bridging is not required unless an additional thromboembolic risk factor is present or patient has an older-generation mechanical aortic valve (Ref).

IV: Limited data available: Initial: 18 units/kg/hour (no bolus) starting when INR falls below the therapeutic range; adjust infusion rate to maintain anticoagulation target based on institutional protocol. If patient is to undergo an invasive procedure, discontinue heparin 4 to 6 hours prior to procedure; reinstitute heparin infusion ≥24 hours after the procedure when bleeding risk is acceptable. Continue heparin until warfarin has been reinstituted and INR is within therapeutic range for 2 consecutive days (Ref).

Mechanical heart valve, postsurgical management

Mechanical heart valve, postsurgical management (to transition to warfarin) (off-label use): **Note:** Initiate postoperatively when risk of bleeding is acceptable (Ref).

IV: Limited data available: Initial: 12 to 18 units/kg/hour (no bolus); adjust infusion rate to maintain anticoagulation target based on institutional protocol. Overlap with warfarin until INR is stable and within therapeutic range for ≥ 2 consecutive days (Ref).

Peripheral arterial occlusion, acute

Peripheral arterial occlusion, acute (off-label use): **Note:** Specific dosing information is limited, but anticoagulation is commonly used at the time of diagnosis to limit thrombus propagation while the patient is evaluated for other possible interventions (Ref).

IV: Initial bolus of 60 to 80 units/kg, followed by an initial continuous infusion of 12 to 18 units/kg/hour; adjust infusion rate to maintain anticoagulation target based on institutional protocol (Ref).

Venous thromboembolism, prophylaxis

Venous thromboembolism, prophylaxis (alternative agent): **Note:** Low-weight patients (eg, < 50 kg) may be more sensitive to routine prophylactic doses, increasing the potential for higher than intended levels of anticoagulation; consider adhering to every-12-hour dosing interval (Ref).

Medical patients with acute illness at moderate to high risk for venous thromboembolism: **SUBQ:** 5,000 units every 8 to 12 hours; continue for length of hospitalization or until fully ambulatory (Ref); extended prophylaxis beyond acute hospital stay is not routinely recommended (Ref).

Nonorthopedic surgery:

Patients with active cancer:

SUBQ: 5,000 units 2 to 4 hours *prior to* surgery, then 5,000 units every 8 hours thereafter (Ref) **or** 5,000 units every 8 to 12 hours started ~ 6 to 24 hours *after* surgery (Ref). **Note:** The optimal duration of prophylaxis has not been established, but it is usually given for a minimum of 7 to 10 days; extending for up to 4 weeks may be reasonable in those undergoing major abdominal or pelvic surgery (Ref).

Patients without cancer: **Note:** For patients with moderate or high risk of venous thromboembolism (VTE) and low risk of bleeding.

SUBQ: 5,000 units every 8 to 12 hours, with initial dose given ≥ 2 hours prior to surgery. Alternatively, may postpone pharmacologic prophylaxis until after surgery (eg, high bleeding risk) when it is safe to initiate. Continue until fully ambulatory and risk of VTE has diminished (typically up to 10 days) (Ref).

Orthopedic surgery (eg, hip fracture surgery, total hip arthroplasty, total knee arthroplasty): **SUBQ:** 5,000 units every 8 to 12 hours, with initial dose administered ≥ 12 hours preoperatively or ≥ 12 hours postoperatively once hemostasis is achieved; optimal duration of prophylaxis is unknown, but it is usually given for a minimum of 10 to 14 days and can be extended for up to 35 days; some experts suggest a duration in the lower end of the range (10 to 14 days) for total knee arthroplasty or higher end of the range (~ 30 days) for total hip arthroplasty (Ref). For extended duration of prophylaxis, may transition to an oral anticoagulant or alternative SUBQ anticoagulant with less frequent dosing (Ref).

Pregnancy: **Note:** Dose intensity is individualized based on risk of thrombosis and bleeding complications (Ref).

Prophylactic dose (also referred to as intermediate dose) (Ref):

First trimester: **SUBQ:** 5,000 to 7,500 units every 12 hours (Ref).

Second trimester: **SUBQ:** 7,500 to 10,000 units every 12 hours (Ref).

Third trimester: **SUBQ:** 10,000 units every 12 hours (reduce dose if the aPTT becomes elevated) (Ref).

Adjusted dose (therapeutic): **Note:** For patients at the highest risk of developing VTE (eg, history of recurrent thrombosis, severe thrombophilia), especially those who are receiving long-term therapy with an oral anticoagulant prior to pregnancy (Ref).

SUBQ: 10,000 units every 12 hours; adjust dose to target an aPTT of 1.5 to 2.5 times control, measured 6 hours after injection; monitor aPTT once daily until stable and within therapeutic range, then monitor every 1 to 2 weeks (Ref).

Management around labor and delivery:

Prior to delivery: Discontinue heparin at the onset of spontaneous labor. Prior to planned induction of labor or cesarean delivery, discontinue heparin ≥ 12 hours before in patients receiving 7,500 to 10,000 units every 12 hours or ≥ 24 hours before in patients receiving $>10,000$ units/dose, $>20,000$ units/day, or an adjusted therapeutic dose. Consider checking coagulation parameters before delivery. Appropriate discontinuation is particularly important if neuraxial anesthesia is planned (Ref).

Postpartum : May restart ≥ 4 to 6 hours after vaginal delivery or ≥ 6 to 12 hours after cesarean delivery unless significant bleeding occurred or neuraxial anesthesia was used (Ref). Anticoagulation should continue for up to 6 weeks postpartum, but potentially longer (Ref).

Venous thromboembolism, treatment, deep vein thrombosis and/or pulmonary embolism

Venous thromboembolism, treatment, deep vein thrombosis and/or pulmonary embolism:

Note: IV heparin may be preferred for initial therapy in patients who are hemodynamically unstable, may need invasive procedures or thrombolysis due to extensive clot burden, are obese, have renal failure, or when rapid reversal of anticoagulation may be needed (Ref). If thrombolytics are used, it is recommended to discontinue heparin during administration then resume upon completion of the thrombolytic infusion (Ref).

Inpatient treatment (nonpregnant and pregnant patients): **IV:** Initial: 80 units/kg bolus followed by a continuous infusion of 18 units/kg/hour **or** 5,000 unit bolus followed by 1,333 units/hour; adjust infusion rate to maintain target laboratory values based on institutional protocol (Ref). **Note:** Weight-based dosing is more effective than fixed dosing at reaching therapeutic anticoagulation (Ref).

Example Weight-Based IV Heparin Nomogram for Treatment of Venous Thromboembolism^{a,b}

^aHull 2022a

^bThis is one example of a weight-based heparin dosing nomogram. Each institution should establish their own heparin dosing nomogram. Other heparin nomograms based on aPTT or anti-Factor Xa monitoring may be employed. Therapeutic range for aPTT must be established at each individual laboratory (Dager 2018).

^cUse actual body weight for calculations.

Initial dose and monitoring

→

80 units/kg bolus (maximum dose: 10,000 units)^c, then 18 units/kg/hour (maximum initial infusion: 2,000 units/hour)^c

Obtain aPTT 6 hours after initial heparin bolus

Dosing adjustments and monitoring

If using anti-Factor Xa activity (units/mL)

Response

If using aPTT (seconds)

0 to 0.09

- Bolus 25 units/kg
- Increase infusion by 3 units/kg/hour

- Repeat assay in 6 hours

<40

0.1 to 0.19

- Increase infusion by 2 units/**kg**/hour
- Repeat assay in 6 hours

40 to 49

0.2 to 0.29

- Increase infusion by 1 unit/**kg**/hour
- Repeat assay in 6 hours

50 to 69

0.3 to 0.7

- No change (within therapeutic range)
- Repeat assay in 6 hours
- Once therapeutic for 2 consecutive assays, may change to once-daily assays

70 to 110

0.71 to 0.79

- Decrease infusion by 1 unit/**kg**/hour
- Repeat assay in 6 hours

111 to 120

0.8 to 0.89

- Stop infusion for 1 hour, then decrease by 2 units/**kg**/hour
- Repeat assay 6 hours after restarting the infusion

121 to 130

0.9 to 0.99

- Stop infusion for 1 hour, then decrease by 3 units/**kg**/hour
- Repeat assay 6 hours after restarting the infusion

131 to 140

1 to 1.09

- Stop infusion for 2 hours, then decrease by 4 units/**kg**/hour
- Repeat assay 6 hours after restarting the infusion

141 to 150

≥ 1.1

- Stop infusion for 2 hours, then decrease by 5 units/**kg**/hour and notify clinician
- Repeat assay 6 hours after restarting the infusion

>150

Outpatient treatment : **Note:** Alternative for patients who have a contraindication to other anticoagulants.

Nonpregnant patients: **Note:** Alternative for patients who have a contraindication to other anticoagulants. **SUBQ :** Initial: 333 units/kg, followed by 250 units/kg every 12 hours; this regimen generally does not require dose adjustment for aPTT (Ref).

Pregnant patients (therapeutic, adjusted dose): **SUBQ:** Initial: 10,000 units every 12 hours; adjust dose to target an aPTT of 1.5 to 2.5 times control, measured 6 hours after injection; monitor aPTT once daily until stable and within therapeutic range, then monitor every 1 to 2 weeks (Ref).

Management around labor and delivery:

Prior to delivery: Discontinue heparin at the onset of spontaneous labor. Prior to planned induction of labor or cesarean delivery, discontinue heparin ≥ 24 hours before in patients receiving a therapeutic adjusted dose of heparin. Consider checking coagulation parameters before delivery. Appropriate discontinuation is particularly important if neuraxial anesthesia is planned (Ref).

Postpartum: May restart ≥ 4 to 6 hours after vaginal delivery or ≥ 6 to 12 hours after cesarean delivery, unless significant bleeding occurred or neuraxial anesthesia was used. Optimal duration of anticoagulation is unknown. In general, total duration of anticoagulation (antepartum plus postpartum) should be at least 3 to 6 months with at least 6 weeks postpartum, but potentially longer (Ref).

Transitioning between anticoagulants: Note: This provides general guidance on transitioning between anticoagulants; also refer to local protocol for additional detail.

Transitioning from another anticoagulant to IV heparin:

Transitioning from a therapeutic dose of SUBQ low-molecular-weight heparin or SUBQ fondaparinux to a therapeutic dose of IV heparin: Start IV heparin without a bolus dose (infusion rate depends on the indication) 1 to 2 hours before the next dose of low-molecular-weight heparin (LMWH) or fondaparinux would have been due (Ref).

Transitioning from warfarin to a therapeutic dose of IV heparin: Stop warfarin and, when INR is as close as possible to the lower end of the targeted INR range, start IV heparin without a bolus dose (infusion rate depends on the indication) (Ref).

Transitioning from a direct-acting oral anticoagulant to a therapeutic dose of IV heparin: Stop direct-acting oral anticoagulant (DOAC) and, when the next DOAC dose would have been due, start IV heparin without a bolus dose (infusion rate depends on the indication) (Ref).

Transitioning from IV heparin to another anticoagulant:

Transitioning from a therapeutic dose of IV heparin to therapeutic SUBQ low-molecular-weight heparin or SUBQ fondaparinux: Stop IV heparin and within 1 hour start SUBQ LMWH or SUBQ fondaparinux. **Note:** If aPTT is not within therapeutic range at the time heparin is stopped, consult local protocol (Ref).

Transitioning from a therapeutic dose of IV heparin to warfarin: Start warfarin and continue IV heparin until INR is within therapeutic range (Dager 2018; Hull 2022a). **Note:** Overlap IV heparin with warfarin until INR is ≥ 2 for at least 2 measurements taken ~ 24 hours apart (duration of overlap is ~ 5 days) (Ref).

Transitioning from a therapeutic dose of IV heparin to a direct-acting oral anticoagulant: Start DOAC when the heparin infusion is stopped (consult local protocol if the aPTT is above the target range) (Ref).

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Adult

The renal dosing recommendations are based upon the best available evidence and clinical expertise. Senior Editorial Team: Bruce Mueller, PharmD, FCCP, FASN, FNKF; Jason Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC; Michael Heung, MD, MS.

Altered kidney function: IV, SUBQ: Mild to severe impairment: No initial dosage adjustment necessary; adjust to maintain anticoagulation target based on institutional protocol (Ref).

Renal replacement therapies: Poorly dialyzed (Ref): **IV, SUBQ:** No supplemental dose or initial dosage adjustment necessary in patients receiving renal replacement therapies (eg, hemodialysis, peritoneal dialysis, CRRT, PIRRT); adjust to maintain anticoagulation target based on institutional protocol (Ref).

Dosing: Liver Impairment: Adult

The liver dosing recommendations are based upon the best available evidence and clinical expertise. Senior Editorial Team: Matt Harris, PharmD, MHS, BCPS, FAST, FCCP; Jeong Park, PharmD, MS, BCTXP, FCCP, FAST; Arun Jesudian, MD; Sasan Sakiani, MD.

Note: Alterations in the coagulation cascade (eg, reduced antithrombin III may falsely elevate aPTT and falsely decrease anti-Xa levels) are present in end-stage liver disease, which can cause heparin resistance and impede interpretation of the level of anticoagulation the patient is receiving, necessitating individualized goals and monitoring (Ref).

Liver impairment prior to treatment initiation:

Child-Turcotte-Pugh class A to C: IV, SUBQ: No initial dosage adjustment necessary; adjust to maintain anticoagulation target based on institutional protocol (Ref).

Dosing: Obesity: Adult

The recommendations for dosing in patients with obesity are based upon the best available evidence and clinical expertise. Senior Editorial Team: Jeffrey F. Barletta, PharmD, FCCM; Manjunath P. Pai, PharmD, FCP; Jason A. Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC.

Venous thromboembolism prophylaxis:

Class 1, 2, or 3 obesity (BMI ≥ 30 kg/m²):

SUBQ: 5,000 to 7,500 units every 8 hours (Ref). **Note:** Until large prospective studies are available evaluating dosing and outcomes, it is reasonable to use 5,000 units every 8 hours. In patients with a BMI >50 kg/m², consider 7,500 units every 8 hours (Ref).

Therapeutic anticoagulation, treatment:

Class 1 or 2 obesity (BMI 30 to 39 kg/m²):

IV: Use **actual body weight** for dosing calculation; adjust infusion rate to maintain target laboratory values based on institutional protocol (Ref). Refer to indication-specific dosing recommendations.

Class 3 obesity (BMI ≥ 40 kg/m²):

IV: Use **adjusted body weight** for dosing calculation; adjust infusion rate to maintain target laboratory values based on institutional protocol (Ref). Alternatively, dosing based on **actual body weight** (with or without a bolus) may be considered with an initial reduced maintenance dose (eg, 12 units/kg/hour for VTE treatment) (Ref). Refer to indication-specific dosing recommendations.

Rationale for recommendations:

Heparin is extensively bound to plasma proteins and has a low V_d , which is similar to blood volume. However, increased blood volume is expected with increasing body weight, resulting in increased dosing requirements for patients with obesity (Ref).

Venous thromboembolism prophylaxis: Due to increases in blood volume and likely decreased bioavailability of SUBQ heparin in patients with obesity, higher dosing (7,500 units every 8 hours) has been recommended in those who require venous thromboembolism prophylaxis (Ref). However, a large retrospective cohort study suggests similar occurrence of venous thromboembolism and major bleeding in patients without obesity and those with obesity (BMI ≥ 30 kg/m²) when administered prophylactic unfractionated heparin at 5,000 units every 8 hours (Ref). Data are lacking in patients with a BMI >50 kg/m².

Therapeutic anticoagulation, treatment: Retrospective studies show no difference in time to achievement of therapeutic aPTT or bleeding rates when actual body weight was used for bolus and maintenance treatment dosing without dose capping in the following weight classifications: BMI <30 kg/m²,

BMI 30 to 39.9 kg/m², or BMI ≥40 kg/m² (Ref). In patients with a BMI ≥40 kg/m², a retrospective study using actual body weight for weight-based dosing showed higher likelihood of supratherapeutic aPTT values compared to patients with a BMI <40 kg/m² (Ref).

Dosing: Adjustment for Toxicity: Adult

Heparin-induced thrombocytopenia: Immediately discontinue all heparin therapy and exposures (eg, heparin flushes and heparin-coated catheters) for intermediate probability or high probability heparin-induced thrombocytopenia or heparin-induced thrombocytopenia with thrombosis based on the 4T score; switch to an alternative nonheparin anticoagulant (Ref).

Dosing: Older Adult

Patients >60 years of age may have higher serum levels and clinical response (longer aPTTs) as compared to younger patients receiving similar dosages. Lower dosages may be required.

Dosing: Pediatric

(For additional information see "Heparin (unfractionated): Pediatric drug information")

Dosage guidance:

Safety: Many concentrations of heparin are available and range from 1 to 20,000 units/mL. Carefully examine each prefilled syringe, bag, or vial prior to use to ensure that the correct concentration is chosen. Heparin lock flush solution is intended only to maintain patency of IV devices and is not to be used for anticoagulant therapy.

Central line flush, patency

Central line flush, patency (intermittent doses): Limited data available (Ref): Infants, Children, and Adolescents: When using intermittent flushes of heparin to maintain patency of single and double lumen central catheters, various recommendations exist; refer to institution-specific protocols. Capped polyvinyl chloride catheters and peripheral heparin locks require flushing more frequently (eg, every 6 to 8 hours). Volume of heparin flush is usually similar to volume of catheter (or slightly greater). Dose of heparin flush used should not approach therapeutic unit per kg dose. Additional flushes should be given when stagnant blood is observed in catheter, after catheter is used for drug or blood administration, and after blood withdrawal from catheter.

Congenital heart defect palliated with systemic to pulmonary artery shunts or with high-risk central venous lines, thromboprophylaxis

Congenital heart defect palliated with systemic to pulmonary artery shunts (eg, Sano shunt, Blalock-Taussig shunt, central shunt) or with high-risk central venous lines (eg, previous thrombosis or hypercoagulable states), thromboprophylaxis: Infants, Children, and Adolescents: Low Dose: Continuous IV infusion: 10 to 15 units/kg/hour (Ref).

Extracorporeal membrane oxygenation, anticoagulation

Extracorporeal membrane oxygenation (ECMO) (venoarterial [VA]/cardiac), anticoagulation:

Note: While used to prevent thrombosis, full anticoagulation dosing is necessary.

Infants, Children, and Adolescents: IV: 100 units/kg prior to ECMO cannulation followed by continuous IV heparin infusion to maintain the activated clotting time (ACT) between 180 and 220 seconds; ACT should be checked hourly while patient is on ECMO; additional monitoring targets for heparin therapy are prolongation of the PTT to 1.5 to 2.5 times the control value or an anti-Xa level of 0.3 to 0.7 units/mL (Ref).

Parenteral nutrition additive, venous access patency

Parenteral nutrition (PN) additive, venous access patency: Infants, Children, and Adolescents: 1 unit/mL (final heparin concentration in PN), both central and peripheral. The final concentration of heparin used for PN solutions may need to be decreased to 0.5 units/mL in small infants receiving larger PN volumes in order to avoid approaching therapeutic amounts (Ref).

Peripheral arterial catheters in situ, thromboprophylaxis

Peripheral arterial catheters in situ, thromboprophylaxis: Infants, Children, and Adolescents: Intra-arterial (via arterial catheter): Continuous infusion of heparin at a final concentration of 5 units/mL at 1 mL/hour (Ref).

Thrombosis, treatment

Thrombosis, treatment:

Systemic heparinization:

Infants: IV: Initial loading dose: 75 units/kg over 10 minutes; then initial continuous maintenance infusion at 28 units/kg/hour; adjust dose to maintain an anti-Xa activity of 0.35 to 0.7 units/mL or an aPTT range that correlates to this anti-Xa range or a protamine titration range of 0.2 to 0.4 units/mL (Ref).

Children and Adolescents: IV: Initial loading dose: 75 units/kg over 10 minutes, then initial continuous maintenance infusion at 20 units/kg/hour; adjust dose to maintain an anti-Xa activity of 0.35 to 0.7 units/mL or an aPTT range that correlates to this anti-Xa range or a protamine titration range of 0.2 to 0.4 units/mL (Ref). **Note:** A lower initial continuous maintenance infusion dose of 18 units/kg/hour in older patients has also been suggested (Ref).

Note: Because of variation among hospitals with reagents (lot numbers) and corresponding control of aPTT values, individual institutions should establish unique, institution-specific nomograms based on current reagent. Due to extensive variability within reagents and anti-Xa levels with corresponding aPTTs, a specific nomogram has not been provided; refer to guidelines for a specific nomogram (Ref).

Systemic to pulmonary artery shunt thrombosis (eg, Sano shunt, Blalock-Taussig shunt, central shunt); treatment in patients with congenital heart disease (CHD): Infants, Children, and Adolescents: IV: Bolus: 50 to 100 units/kg, ongoing continuous infusion should be considered (Ref).

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Pediatric

All patients: No dosage adjustment required; adjust therapeutic heparin according to aPTT or anti-Xa activity

Dosing: Liver Impairment: Pediatric

All patients: No dosage adjustment required; adjust therapeutic heparin according to aPTT or anti-Xa activity

Adverse Reactions (Significant): Considerations

Major bleeding

Heparin increases the risk of bleeding (**hemorrhage**), including major bleeding. Bleeding may occur at any site and may be severe enough to be life-threatening, with reports of fatal cases (eg, intracranial bleeding) (Ref). Bleeding is reversible with the use of protamine.

Mechanism: Dose-related; binds to antithrombin III, increasing its activity 1,000-fold (Ref). This complex inactivates factors IIa (thrombin), Xa, IXa, XIa, and XIIa, leading to decreased coagulation (Ref).

Onset: Varied; may occur at any time but data show that it is more common toward the end of planned therapy (Ref). Bleeding may occur sooner; however, this is largely driven by other risk factors for bleeding that are present (Ref).

Risk factors:

- Supratherapeutic dosing (excess bolus or infusion) or aPTT level (Ref)
- Kidney impairment (Ref)
- Uncontrolled and severe hypertension

- History of hemorrhagic stroke
- Recent gastrointestinal bleed
- Females
- Age >60 years
- Concurrent antiplatelet therapy (eg, P2Y₁₂ inhibitors, GPIIb/IIIa inhibitors) (Ref)
- Concurrent use of thrombolytics (Ref)
- Liver disease with impaired hemostasis
- Ulcerative GI lesions
- Continuous GI tube drainage
- Subacute bacterial endocarditis
- Concurrent coagulation disorders (Ref)
- Patients with hereditary antithrombin III deficiency receiving concurrent antithrombin III therapy
- Recent brain, spinal, or ophthalmic surgery
- Recent surgery, trauma, or invasive procedures (Ref)
- Spinal interventions, including epidurals or spinal taps (Ref)

Osteoporosis

Studies primarily in patients who are pregnant show that heparin may cause significant **decreased bone mineral density** (BMD) $\geq 10\%$ decrease from baseline femur measurements (Ref). A study in patients on hemodialysis showed decreases in BMD primarily in the lumbar spine (Ref). BMD losses may still be significant 6 months after discontinuation (Ref). **Bone fractures** are uncommon (Ref).

Mechanism: Dose- and time-related (Ref); binds to osteoprotegerin, sterically inhibiting its interaction with RANKL on osteoblasts. This promotes RANK-RANKL linkages, activating osteoclasts and causing bone resorption (Ref).

Onset: Delayed; occurs after about 3 to 6 months (Ref).

Risk factors:

- High doses (15,000 to 30,000 units/day), but this is controversial (Ref)
- Long-term use of at least 3 to 6 months (Ref)
- Use of unfractionated heparin (versus low-molecular weight heparin) (Ref)
- Pregnancy (Ref)

Thrombocytopenia

Mild **thrombocytopenia** may occur during therapy (Ref). **Heparin-induced thrombocytopenia** without thrombosis (HIT) or with thrombosis (HITT) may also occur, warranting discontinuation of therapy, and can be fatal without treatment (Ref). HIT/HITT should be suspected if there is a 30% to 50% reduction in platelet count from baseline with or without new thrombosis formation while on heparin therapy (Ref). The 4T score can be used to evaluate the probability of HIT and whether to further pursue laboratory evaluation (Ref). HIT/HITT antibodies resolve approximately 90 to 100 days after discontinuation of therapy; however, it is unknown if those with a history of HIT/HITT are more likely to develop the reaction again if re-exposed to heparin. Therefore, avoidance of heparin is recommended (or allowance of only short-term use for specific indications) even after resolution of antibodies (Ref).

Mechanism: Time-related; HIT is an immune-mediated complication where IgG antibodies directed at complexes of heparin and the platelet protein, platelet factor 4 (PF4), bind to the Fcγ1a receptors on platelets (Ref). This causes platelet activation, aggregation, and significant thrombin generation,

leading to thrombosis (Ref). Thrombocytopenia is secondary to the significant activation and usage of platelets, effectively removing them from circulation (Ref).

Onset: Varied; the decrease in platelet count typically occurs 5 to 10 days after heparin initiation; however, platelet count can fall within 24 hours after heparin initiation in “rapid-onset HIT,” which tends to occur in individuals with circulating heparin antibodies secondary to a recent exposure (Ref). “Delayed-onset HIT” is also possible, which occurs when platelet count fall occurs weeks after heparin is discontinued (Ref). Thrombosis can precede thrombocytopenia in up to 25% of cases (Ref).

Risk factors for HIT:

- Longer courses of heparin (Ref)
- Use of unfractionated heparin (versus low molecular weight heparin) (Ref)
- Bovine heparin (Ref)
- Patients undergoing surgery (particularly cardiac and orthopedic surgery) (Ref)
- Patients presenting with major trauma (Ref)
- Females (Ref)

Additional risk factors for HIT:

- High antibody levels (optical density or titer) (Ref)
- Marked thrombocytopenia (>90% decline from baseline) (Ref)

Adverse Reactions

The following adverse drug reactions are derived from product labeling unless otherwise specified.

Postmarketing:

Cardiovascular: Cardiac tamponade (Su 2005), vasospasm

Dermatologic: Transient alopecia

Endocrine & metabolic: Hyperkalemia (Su 2005), suppression of aldosterone synthesis

Genitourinary: Priapism (Burke 1983)

Hematologic & oncologic: Hemorrhage (including adrenal hemorrhage, ovarian hemorrhage, retroperitoneal hemorrhage) (Krishnaswamy 2011), heparin-induced thrombocytopenia, thrombocytopenia, thrombosis in heparin-induced thrombocytopenia (including acute myocardial infarction, cerebral thrombosis, cerebrovascular accident, deep vein thrombosis, mesenteric thrombosis, peripheral gangrene, pulmonary embolism, renal artery thrombosis, skin necrosis)

Hepatic: Increased serum alanine aminotransferase, increased serum aspartate aminotransferase

Hypersensitivity: Anaphylactic shock (Bottio 2003), hypersensitivity reaction (including pruritus and burning sensation of feet [plantar side]), infusion-related reaction (skin necrosis), nonimmune anaphylaxis

Local: Erythema at injection site (SUBQ), hematoma at injection site (SUBQ), irritation at injection site (SUBQ), pain at injection site (SUBQ), skin ulceration at injection site (SUBQ), tissue necrosis at injection site (SUBQ)

Neuromuscular & skeletal: Bone fracture (Barbour 1994), decreased bone mineral density (Douketis 1996), osteoporosis (with long-term use) (Casele 2006)

Contraindications

Hypersensitivity to heparin or any component of the formulation (unless a life-threatening situation necessitates use and use of an alternative anticoagulant is not possible); severe thrombocytopenia; history of heparin-induced thrombocytopenia; history of heparin-induced thrombocytopenia with

thrombosis; uncontrolled active bleeding (except when this is due to disseminated intravascular coagulation); not for use when appropriate blood coagulation tests cannot be obtained at appropriate intervals (applies to full-dose heparin only).

Note: Some products contain benzyl alcohol as a preservative; their use in neonates, infants, or pregnant or breastfeeding patients is contraindicated by some manufacturers.

Warnings/Precautions

Concerns related to adverse effects:

- Heparin resistance: Dose requirements >35,000 units/24 hours to maintain a therapeutic aPTT may occur in patients with antithrombin deficiency, increased heparin clearance, elevations in heparin-binding proteins, and elevations in factor VIII and/or fibrinogen; frequently encountered in patients with fever, thrombosis, thrombophlebitis, infections with thrombosing tendencies, myocardial infarction, cancer, and in postsurgical patients; measurement of anticoagulant effects using anti-Factor Xa levels may be of benefit.
- Hepatic effects: Elevations in serum aminotransferases have been observed during therapy. These elevations should be evaluated with caution as they may occur and resolve in the setting of the underlying condition for which heparin is being used.
- Hypersensitivity reactions: Hypersensitivity reactions, including fever, chills, urticaria, asthma, rhinitis, lacrimation, and anaphylaxis, have been reported. In patients with a documented hypersensitivity reaction, heparin should only be considered in life-threatening situations when use of an alternative anticoagulant is not possible. Some products are derived from animal tissue and may be contraindicated in patients with animal allergies (ie, pork); consult individual prescribing information.

Dosage form specific issues:

- Benzyl alcohol and derivatives: Some dosage forms may contain benzyl alcohol as a preservative. In neonates, large amounts of benzyl alcohol (≥ 99 mg/kg/day) have been associated with a potentially fatal toxicity ("gasping syndrome"); the "gasping syndrome" consists of metabolic acidosis, respiratory distress, gasping respirations, CNS dysfunction (including convulsions, intracranial hemorrhage), hypotension, and cardiovascular collapse (AAP ["Inactive" 1997]; CDC 1982); some data suggests that benzoate displaces bilirubin from protein binding sites (Ahlfors 2001); avoid or use dosage forms containing benzyl alcohol with caution in neonates. See manufacturer's labeling. Use in neonates, infants, or pregnant or nursing mothers is contraindicated by some manufacturers; the use of preservative-free heparin is, therefore, recommended in these populations.
- Sulfites: Some preparations contain sulfite which may cause allergic reactions.

Other warnings/precautions:

- Fatal medication errors: Many concentrations of heparin are available ranging from 1 unit/mL to 20,000 units/mL. Clinicians **must** carefully examine each prefilled syringe or vial prior to use ensuring that the correct concentration is chosen; fatal hemorrhages have occurred related to heparin overdose especially in pediatric patients.

Warnings: Additional Pediatric Considerations

Confirm the concentration of all heparin injection vials prior to administration; do not use heparin injection as a "catheter lock flush" as the injection is supplied in various concentrations including highly concentrated strengths. Fatal hemorrhages have occurred in pediatric patients when higher concentrations of heparin injection were confused with lower concentrations of heparin lock flush.

Heparin-induced thrombocytopenia (HIT) has been reported in pediatric patients; incidence and risk factors are not well-defined due to variability related to patient inclusion and laboratory techniques; incidence rates up to 2.3% in PICU patients have been reported, and cases observed at both low and high levels of heparin exposure (eg, venous access device line flushes and supratherapeutic doses associated with hemodialysis and during cardiopulmonary bypass). Monitor platelet count closely; if HIT develops, consider alternate anticoagulation therapy (eg, danaparoid, argatroban) (ACCP [Monagle 2012]).

Heparin resistance should be suspected in pediatric patients if unable to achieve an activated clotting time (ACT) >300 seconds after administration of >600 units/kg (ACCP [Giglia 2013]).

Dosage Forms: US

Excipient information presented when available (limited, particularly for generics); consult specific product labeling. [DSC] = Discontinued product

Solution, Intravenous:

Generic: 2000 units (1000 mL); 25,000 units (500 mL); 2000 units in 0.9% NaCl per liter (1000 mL); 25,000 units/500 mL (500 mL); 25,000 units/500 mL in NaCl 0.45% (500 mL)

Solution, Intravenous [preservative free]:

Generic: 2000 units in 0.9% NaCl per liter (1000 mL); 25,000 units/500 mL in NaCl 0.45% (500 mL)

Solution, Injection, as sodium:

Generic: 1000 units/mL (1 mL, 10 mL, 30 mL); 5000 units/mL (1 mL, 10 mL); 5000 units/0.5 mL (0.5 mL); 10,000 units/mL (1 mL, 4 mL, 5 mL); 20,000 units/mL (1 mL)

Solution, Injection, as sodium [preservative free]:

Generic: 1000 units/mL (2 mL); 5000 units/mL (1 mL); 5000 units/0.5 mL (0.5 mL)

Solution, Intravenous, as sodium:

Generic: 1000 units (500 mL); 10,000 units (250 mL [DSC]); 12,500 units (250 mL [DSC]); 20,000 units (500 mL); 25,000 units (250 mL); 1 units/mL (1 mL, 2 mL, 2.5 mL, 3 mL, 5 mL, 10 mL); 10 units/mL (1 mL, 5 mL, 10 mL); 100 units/mL (1 mL, 2 mL, 2.5 mL, 3 mL, 5 mL, 10 mL); 1000 units/500 mL in NaCl 0.9% (500 mL); 12,500 units/250 mL in NaCl 0.45% (250 mL); 25,000 units/250 mL in Dextrose 5% (250 mL); 25,000 units/250 mL in NaCl 0.45% (250 mL)

Solution, Intravenous, as sodium [preservative free]:

Generic: 1000 units (500 mL); 10 units/mL (1 mL, 2 mL, 2.5 mL, 3 mL, 5 mL, 10 mL); 25,000 units/250 mL in Dextrose 5% (250 mL); 25,000 units/250 mL in NaCl 0.45% (250 mL)

Solution Prefilled Syringe, Injection, as sodium [preservative free]:

Generic: 5000 units/0.5 mL (0.5 mL)

Generic Equivalent Available: US

Yes

Pricing: US

Solution (Heparin (Porcine) in NaCl Intravenous)

1000UT/500ML 0.9% (per mL): \$0.01 - \$0.02

2000UNIT/L 0.9% (per mL): \$0.01

12500UT/250ML 0.45% (per mL): \$0.24

25000UT/250ML 0.45% (per mL): \$0.05 - \$0.08

25000UT/500ML 0.45% (per mL): \$0.02 - \$0.04

Solution (Heparin Na (Pork) Lock Flush PF Intravenous)

1 units/mL (per mL): \$1.16 - \$1.25

10 units/mL (per mL): \$1.00 - \$1.10

100 units/mL (per mL): \$1.03 - \$1.13

Solution (Heparin Sod (Porcine) in D5W Intravenous)

40UNIT/ML 5% (per mL): \$0.03

100 units/mL (per mL): \$0.05 - \$0.08

25000UT/500ML 5% (per mL): \$0.03 - \$0.04

Solution (Heparin Sod (Pork) Lock Flush Intravenous)

10 units/mL (per mL): \$0.13

100 units/mL (per mL): \$0.21

Solution (Heparin Sodium (Porcine) Injection)

1000 units/mL (per mL): \$0.75 - \$4.64

5000 units/mL (per mL): \$0.83 - \$5.40

10000 units/mL (per mL): \$1.39 - \$10.44

20000 units/mL (per mL): \$12.00 - \$20.11

Solution (Heparin Sodium (Porcine) PF Injection)

1000 units/mL (per mL): \$1.69 - \$7.44

5000 units/0.5 mL (per 0.5 mL): \$4.96 - \$7.20

5000 units/mL (per mL): \$7.20

Solution Prefilled Syringe (Heparin Sodium (Porcine) Injection)

5000 units/0.5 mL (per 0.5 mL): \$3.72 - \$3.72

Disclaimer: A representative AWP (Average Wholesale Price) price or price range is provided as reference price only. A range is provided when more than one manufacturer's AWP price is available and uses the low and high price reported by the manufacturers to determine the range. The pricing data should be used for benchmarking purposes only, and as such should not be used alone to set or adjudicate any prices for reimbursement or purchasing functions or considered to be an exact price for a single product and/or manufacturer. Medi-Span expressly disclaims all warranties of any kind or nature, whether express or implied, and assumes no liability with respect to accuracy of price or price range data published in its solutions. In no event shall Medi-Span be liable for special, indirect, incidental, or consequential damages arising from use of price or price range data. Pricing data is updated monthly.

Dosage Forms: Canada

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Solution, Intravenous:

Generic: 25,000 units/500 mL (500 mL)

Solution, Injection, as sodium:

Generic: 1000 units/mL (1 mL, 2 mL, 10 mL, 30 mL); 5000 units/0.5 mL (0.5 mL); 10,000 units/mL (0.5 mL, 1 mL, 5 mL); 1000 units (500 mL, 1000 mL)

Solution, Intravenous, as sodium:

Generic: 20,000 units/500 mL in Dextrose 5% (500 mL); 100 units/mL (2 mL, 10 mL); 25,000 units/250 mL in Dextrose 5% (250 mL)

Solution Prefilled Syringe, Subcutaneous:

Generic: 5000 units/0.5 mL (0.5 mL)

Administration: Adult

SUBQ: Inject in subcutaneous tissue only (not muscle tissue). Injection sites should be rotated (usually left and right portions of the abdomen, above iliac crest).

IM: Do not administer IM due to pain, irritation, and hematoma formation.

Inhalation: Treatment of burn inhalation injury (off-label route): Administer via mesh nebulizer (Ref). Refer to institution-specific protocols.

Continuous IV infusion: Infuse via infusion pump. If preparing solution, mix thoroughly prior to administration.

Catheter patency: Practice is variable. If used for this purpose, refer to institution-specific protocols. The intention is only to maintain patency of IV catheters and is **not** to be used for anticoagulant therapy.

Intravesical (off-label use): Various dosage regimens of heparin (20,000 to 50,000 units) alone or with alkalinized lidocaine (1% to 4%) have been instilled into the bladder.

Intra-arterial (off-label use): Infuse via infusion pump. If preparing solution, mix thoroughly prior to administration (Ref).

Preparation for Administration: Adult

Inhalation: Dilute 5,000 to 10,000 units/mL heparin solution with isotonic saline to yield a final concentration of 5,000 to 10,000 units of heparin per 3 mL (Ref). Refer to institution-specific protocols.

Usual Infusion Concentrations: Adult

Note: Premixed solutions available

IV infusion: 25,000 units in 250 mL (concentration: 100 units/mL) of D₅W, 1/2 NS, or NS

Administration: Pediatric

Note: Many concentrations of heparin are available and range from 1 to 20,000 units/mL. Carefully examine each prefilled syringe, bag, or vial prior to use to ensure that the correct concentration is chosen.

Parenteral: Do **not** administer IM due to pain, irritation, and hematoma formation.

IV:

IV bolus: Administered over 10 minutes (Ref).

Continuous IV infusion: Infuse via infusion pump.

Heparin lock: Inject via injection cap using positive pressure flushing technique. Heparin lock flush solution is intended only to maintain patency of IV devices and is **not** to be used for anticoagulant therapy.

Central venous catheters: Must be flushed with heparin solution when newly inserted, daily (at the time of tubing change), after blood withdrawal or transfusion, and after an intermittent infusion through an injectable cap.

SubQ: Not all preparation intended for SubQ administration, verify product selection. Inject in subcutaneous tissue only (not muscle tissue). Injection sites should be rotated (usually left and right portions of the abdomen, above iliac crest).

Preparation for Administration: Pediatric

Note: Many concentrations of heparin are available and range from 1 to 20,000 units/mL. Carefully examine each prefilled syringe, bag, or vial prior to use to ensure that the correct concentration is chosen. Heparin lock flush solution is intended only to maintain patency of IV devices and is not to be used for anticoagulant therapy.

Parenteral: Concentration should be determined based on indication and dose to allow for safe and accurate administration of the correct dose. After addition of heparin to the infusion solution, invert the solution at least 6 times to ensure adequate mixing and prevent pooling of heparin. Products containing benzyl alcohol or derivatives should not be used to prepare products for neonates.

Vermont Oxford Network recommends a standard concentration of 0.5 units/mL (in 0.45% NS) for maintaining line patency in neonates (Ref). ASHP recommends standard concentrations of 50 units/mL and 100 units/mL for anticoagulation therapy and 0.5 units/mL (neonatal) and 2 units/mL for arterial line maintenance (Ref).

Usual Infusion Concentrations: Pediatric

Note: Premixed solutions available

IV infusion: 100 units/mL

Storage/Stability

Heparin solutions are colorless to slightly yellow. Minor color variations do not affect therapeutic efficacy. Heparin should be stored at room temperature. Protect from freezing and temperatures >40°C.

Stability at room temperature and refrigeration:

Prepared bag: Variable (specific to solution, concentration, and/or study conditions); also refer to manufacturer's labeling.

Premixed bag: After seal is broken, 4 days; also refer to manufacturer's labeling.

Out of overwrap stability: 30 days; also refer to manufacturer's labeling.

After diluting with isotonic saline for burn inhalation injury: Store 10 days under refrigeration at 2°C to 8°C (36°F to 46°F) or 4 days at 20°C to 25°C (68°F to 77°F) (Ref).

Use: Labeled Indications

Anticoagulation: Prophylaxis and/or treatment of thromboembolic disorders (eg, venous thromboembolism, pulmonary embolism) and thromboembolic complications associated with atrial fibrillation or other disease states; prevention of clotting in arterial and cardiac surgery; as an anticoagulant for blood transfusions, extracorporeal circulation, and dialysis procedures.

Note: Heparin lock flush solution and certain premixed infusion bags (eg, 2 units/mL) are intended only to maintain patency of IV devices and are **not** to be used for systemic anticoagulant therapy.

Use: Off-Label: Adult

Antibiotic lock technique, adjunctive therapy (catheter-salvage strategy); Burn inhalation injury; Extracorporeal membrane oxygenation, anticoagulation of circuit; Frostbite; Mechanical heart valve, bridging anticoagulation (for interruptions in warfarin therapy); Mechanical heart valve, postsurgical management (to transition to warfarin); Non-ST-elevation acute coronary syndromes; Percutaneous coronary intervention; Peripheral arterial occlusion, acute; ST-elevation myocardial infarction

Medication Safety Issues

Sound-alike/look-alike issues:

Heparin may be confused with Hespán

High alert medication:

The Institute for Safe Medication Practices (ISMP) includes this medication among its list of drug classes (anticoagulants, parenteral and oral) which have a heightened risk of causing significant patient harm when used in error (ISMP List of High-Alert Medications in Acute Care Settings ©ISMP 2024; ISMP List of High-Alert Medications in Community/Ambulatory Care Settings ©ISMP 2021; High-Alert Medications in Long-Term Care (LTC) Settings ©ISMP 2021).

National Patient Safety Goals:

The Joint Commission (TJC) requires healthcare organizations that provide anticoagulant therapy to have approved protocols and evidence-based practice guidelines in place to reduce the risk of

anticoagulant-associated patient harm. Patients receiving anticoagulants should receive individualized care through a defined process that includes medication selection, dosing (including adjustments for age, renal function, or liver function), drug-drug interactions, drug-food interactions, other applicable risk factors, monitoring, patient and family education, proper administration, reversal of anticoagulation, management of bleeding events, and perioperative management. This does not apply to routine short-term use of anticoagulants for prevention of venous thromboembolism during procedures or hospitalizations (NPSG.03.05.01).

Older Adult: High-Risk Medication:

Heparin is identified in the Screening Tool of Older Person's Prescriptions (STOPP) criteria as a potentially inappropriate medication in older adults (≥ 65 years of age) with a history of gastric antral vascular ectasia (O'Mahony 2023).

Administration issues:

The 100 unit/mL concentration should not be used to flush heparin locks, IV lines, or intra-arterial lines in neonates or infants < 10 kg (systemic anticoagulation may occur). The 10 unit/mL flush concentration may inadvertently cause systemic anticoagulation in infants < 1 kg who receive frequent flushes.

Other safety concerns:

Heparin sodium injection 10,000 units/mL and Hep-Lock U/P 10 units/mL have been confused with each other. Fatal medication errors have occurred between the two whose labels are both blue. **Never rely on color as a sole indicator to differentiate product identity.**

Heparin lock flush solution is intended only to maintain patency of IV devices and is **not** to be used for anticoagulant therapy.

Metabolism/Transport Effects

None known.

Drug Interactions

(For additional information: Launch drug interactions program)

Note: Interacting drugs may **not be individually listed below** if they are part of a group interaction (eg, individual drugs within "CYP3A4 Inducers [Strong]" are NOT listed). For a complete list of drug interactions by individual drug name and detailed management recommendations, use the drug interactions program by clicking on the "Launch drug interactions program" link above.

Acalabrutinib: May increase anticoagulant effects of Anticoagulants. *Risk C: Monitor*

Aducanumab: May increase anticoagulant effects of Anticoagulants. Management: Use caution and monitor patients for signs and symptoms of bleeding if these agents are combined. Additionally, the Alzheimer's Disease and Related Disorders Therapeutic Work Group recommends against concomitant use of anticoagulants and aducanumab. *Risk D: Consider Therapy Modification*

Aescin: May increase anticoagulant effects of Anticoagulants. Management: Consider alternatives to this combination as aescin product labeling states coadministration with anticoagulants is not recommended. *Risk D: Consider Therapy Modification*

Alemtuzumab: May increase anticoagulant effects of Anticoagulants. *Risk C: Monitor*

Alteplase: May increase anticoagulant effects of Anticoagulants. *Risk X: Avoid*

Anacaulase: May increase anticoagulant effects of Anticoagulants. *Risk C: Monitor*

Anagrelide: May increase anticoagulant effects of Anticoagulants. *Risk C: Monitor*

Andexanet Alfa (Coagulation Factor Xa [Recombinant], Inactivated): May decrease therapeutic effects of Heparin. *Risk X: Avoid*

Angiotensin II Receptor Blockers: Heparin may increase hyperkalemic effects of Angiotensin II Receptor Blockers. *Risk C: Monitor*

Angiotensin-Converting Enzyme Inhibitors: Heparin may increase hyperkalemic effects of Angiotensin-Converting Enzyme Inhibitors. *Risk C: Monitor*

Anticoagulants (Miscellaneous Agents): Anticoagulant effects of Anticoagulants (Miscellaneous Agents) and Heparin may increase with coadministration. Management: Discontinue other agents that can increase the risk of hemorrhage, such as heparin, prior to initiating therapy with these anticoagulants. If combined, monitor closely for signs and symptoms of bleeding. *Risk D: Consider Therapy Modification*

Antidepressants with Antiplatelet Effects: May increase anticoagulant effects of Heparin. *Risk C: Monitor*

Antiplatelet Agents (P2Y₁₂ Inhibitors): May increase anticoagulant effects of Heparin. *Risk C: Monitor*

Antithrombin: May increase anticoagulant effects of Heparin. *Risk C: Monitor*

Apixaban: May increase anticoagulant effects of Anticoagulants. Refer to separate drug interaction content and to full drug monograph content regarding use of apixaban with vitamin K antagonists (eg, warfarin, acenocoumarol) during anticoagulant transition and bridging periods. *Risk X: Avoid*

Aspirin: May increase anticoagulant effects of Heparin. *Risk C: Monitor*

Bromperidol: May increase adverse/toxic effects of Anticoagulants. *Risk C: Monitor*

Caplacizumab: May increase anticoagulant effects of Anticoagulants. Management: Avoid coadministration of caplacizumab with antiplatelets if possible. If coadministration is required, monitor closely for signs and symptoms of bleeding. Interrupt use of caplacizumab if clinically significant bleeding occurs. *Risk D: Consider Therapy Modification*

Chloroprocaine (Systemic): Anticoagulants may increase adverse/toxic effects of Chloroprocaine (Systemic). *Risk C: Monitor*

Cilostazol: May increase anticoagulant effects of Anticoagulants. *Risk C: Monitor*

Collagenase (Systemic): Anticoagulants may increase adverse/toxic effects of Collagenase (Systemic). Specifically, the risk of injection site bruising and/or bleeding may be increased. *Risk C: Monitor*

Dabigatran Etexilate: May increase anticoagulant effects of Anticoagulants. Refer to separate drug interaction content and to full drug monograph content regarding use of dabigatran etexilate with vitamin K antagonists (eg, warfarin, acenocoumarol) during anticoagulant transition and bridging periods. *Risk X: Avoid*

Dasatinib: May increase anticoagulant effects of Anticoagulants. *Risk C: Monitor*

Deferasirox: Anticoagulants may increase adverse/toxic effects of Deferasirox. Specifically, the risk for GI ulceration/irritation or GI bleeding may be increased. *Risk C: Monitor*

Defibrotide: May increase anticoagulant effects of Anticoagulants. *Risk X: Avoid*

Deoxycholic Acid: Anticoagulants may increase adverse/toxic effects of Deoxycholic Acid. Specifically, the risk for bleeding or bruising in the treatment area may be increased. *Risk C: Monitor*

Dipyridamole: May increase adverse/toxic effects of Heparin. Specifically, the risk of bleeding may be increased. Management: Use caution and reduce the dose of heparin or dipyridamole if these agents are combined. *Risk D: Consider Therapy Modification*

Donanemab: May increase anticoagulant effects of Anticoagulants. Management: Use caution and monitor patients for signs and symptoms of bleeding if these agents are combined. The Alzheimer's Disease and Related Disorders Therapeutic Work Group recommends against concomitant use of anticoagulants and donanemab. *Risk D: Consider Therapy Modification*

Drospirenone-Containing Products: May increase hyperkalemic effects of Heparin. *Risk C: Monitor*

Edoxaban: May increase anticoagulant effects of Anticoagulants. Refer to separate drug interaction content and to full drug monograph content regarding use of edoxaban with vitamin K antagonists (eg, warfarin, acenocoumarol) during anticoagulant transition and bridging periods. Management: Some limited combined use may be indicated during periods of transition from one anticoagulant to another. See the full edoxaban drug monograph for specific recommendations on switching anticoagulant treatment. *Risk X: Avoid*

Factor X (Human): Anticoagulants (Inhibitors of Factor Xa) may decrease therapeutic effects of Factor X (Human). *Risk C: Monitor*

Fondaparinux: Anticoagulant effects of Anticoagulants and Fondaparinux may increase with coadministration. Management: Discontinue other anticoagulants prior to initiating fondaparinux if possible. If coadministration is required, monitor closely for signs and symptoms of hemorrhage. *Risk D: Consider Therapy Modification*

Glycoprotein IIb/IIIa Inhibitors: May increase anticoagulant effects of Heparin. *Risk C: Monitor*

Hemin: May increase anticoagulant effects of Anticoagulants. *Risk X: Avoid*

Heparins (Low Molecular Weight): Anticoagulant effects of Heparin and Heparins (Low Molecular Weight) may increase with coadministration. Management: Discontinue other agents that can increase the risk of hemorrhage, such as heparin prior to initiating therapy with low molecular weight heparins. If coadministration is required, monitor closely for signs and symptoms of bleeding. *Risk D: Consider Therapy Modification*

Herbal Products with Anticoagulant/Antiplatelet Effects: May increase adverse/toxic effects of Anticoagulants. Bleeding may occur. *Risk C: Monitor*

Ibritumomab Tiuxetan: Anticoagulants may increase adverse/toxic effects of Ibritumomab Tiuxetan. Both agents may contribute to an increased risk of bleeding. *Risk C: Monitor*

Ibrutinib: Anticoagulants may increase adverse/toxic effects of Ibrutinib. Specifically, the risks of bleeding and hemorrhage may be increased. *Risk C: Monitor*

Icosapent Ethyl: May increase anticoagulant effects of Anticoagulants. *Risk C: Monitor*

Inotersen: May increase anticoagulant effects of Anticoagulants. *Risk C: Monitor*

Kanamycin: May increase anticoagulant effects of Anticoagulants. *Risk C: Monitor*

Lecanemab: May increase adverse/toxic effects of Anticoagulants. Specifically, the risk of hemorrhage may be increased. Management: Monitor patients for signs and symptoms of bleeding if these agents are combined. The Alzheimer's Disease and Related Disorders Therapeutic Work Group recommends against concomitant use and this combination is contraindicated in labels outside of the US. *Risk D: Consider Therapy Modification*

Levothyroxine: Heparin may decrease serum concentrations of Levothyroxine. *Risk C: Monitor*

Limaprost: May increase adverse/toxic effects of Anticoagulants. The risk for bleeding may be increased. *Risk C: Monitor*

Lipid Emulsion (Fish Oil Based): May increase anticoagulant effects of Anticoagulants. Lipid Emulsion (Fish Oil Based) may decrease anticoagulant effects of Anticoagulants. *Risk C: Monitor*

Lysozyme: May decrease therapeutic effects of Heparin. *Risk C: Monitor*

Mesoglycan: May increase anticoagulant effects of Anticoagulants. *Risk C: Monitor*

MiFEPRIStone: May increase adverse/toxic effects of Anticoagulants. Specifically, the risk of bleeding may be increased. *Risk X: Avoid*

Miscellaneous Antiplatelets: May increase anticoagulant effects of Heparin. Management: Reduce the dose of heparin or these antiplatelets if these agents are coadministered to reduce the risk of bleeding. *Risk D: Consider Therapy Modification*

Nintedanib: Anticoagulants may increase adverse/toxic effects of Nintedanib. Specifically, the risk for bleeding may be increased. *Risk C: Monitor*

Nitroglycerin: May decrease anticoagulant effects of Heparin. Nitroglycerin may decrease serum concentrations of Heparin. *Risk C: Monitor*

Nonsteroidal Anti-Inflammatory Agents (COX-2 Selective): May increase anticoagulant effects of Anticoagulants. *Risk C: Monitor*

Nonsteroidal Anti-Inflammatory Agents (Nonselective): May increase anticoagulant effects of Heparin. *Risk C: Monitor*

Nonsteroidal Anti-Inflammatory Agents (Ophthalmic): May increase anticoagulant effects of Anticoagulants. *Risk C: Monitor*

Nonsteroidal Anti-Inflammatory Agents (Topical): May increase anticoagulant effects of Anticoagulants. *Risk C: Monitor*

Obinutuzumab: Anticoagulants may increase adverse/toxic effects of Obinutuzumab. Specifically, the risk of serious bleeding-related events may be increased. Management: Consider avoiding coadministration of obinutuzumab and anticoagulants, especially during the first cycle of obinutuzumab therapy. *Risk D: Consider Therapy Modification*

Omacetaxine: Anticoagulants may increase adverse/toxic effects of Omacetaxine. Specifically, the risk for bleeding-related events may be increased. Management: Avoid concurrent use of anticoagulants with omacetaxine in patients with a platelet count of less than 50,000/uL. *Risk X: Avoid*

Omega-3 Fatty Acids: May increase anticoagulant effects of Anticoagulants. *Risk C: Monitor*

Oritavancin: May decrease therapeutic effects of Heparin. Specifically, oritavancin may artificially increase the results of laboratory tests commonly used to monitor IV heparin effectiveness, which could lead to incorrect decisions to decrease heparin doses. *Risk X: Avoid*

Ozagrel: May increase anticoagulant effects of Anticoagulants. Management: Avoid coadministration of ozagrel and anticoagulants if possible. If coadministration is required, use caution, monitor patients closely for signs and symptoms of bleeding, and consider ozagrel or anticoagulant dose reductions. *Risk D: Consider Therapy Modification*

Pentosan Polysulfate Sodium: May increase anticoagulant effects of Anticoagulants. *Risk C: Monitor*

Pentoxifylline: May increase anticoagulant effects of Heparin. *Risk C: Monitor*

Pirtobrutinib: May increase anticoagulant effects of Anticoagulants. *Risk C: Monitor*

Potassium Salts: Heparin may increase hyperkalemic effects of Potassium Salts. *Risk C: Monitor*

Potassium-Sparing Diuretics: Heparin may increase hyperkalemic effects of Potassium-Sparing Diuretics. *Risk C: Monitor*

Protein C Concentrate (Human): May increase anticoagulant effects of Anticoagulants. *Risk C: Monitor*

Remibrutinib: Anticoagulants may increase adverse/toxic effects of Remibrutinib. Specifically, the risk for bleeding may be increased. *Risk C: Monitor*

Retepase: May increase anticoagulant effects of Anticoagulants. *Risk C: Monitor*

Rivaroxaban: Anticoagulants may increase anticoagulant effects of Rivaroxaban. Refer to separate drug interaction content and to full drug monograph content regarding use of rivaroxaban with vitamin K antagonists (eg, warfarin, acenocoumarol) during anticoagulant transition and bridging periods. *Risk X: Avoid*

Salicylates: May increase anticoagulant effects of Anticoagulants. *Risk C: Monitor*

Streptokinase: May increase anticoagulant effects of Anticoagulants. *Risk X: Avoid*

Sugammadex: May increase anticoagulant effects of Anticoagulants. *Risk C: Monitor*

Sulodexide: May increase anticoagulant effects of Anticoagulants. *Risk C: Monitor*

Telavancin: May decrease therapeutic effects of Heparin. Specifically, telavancin may artificially increase the results of laboratory tests commonly used to monitor IV heparin effectiveness, which could lead to incorrect decisions to decrease heparin doses. *Risk X: Avoid*

Tenecteplase: May increase anticoagulant effects of Anticoagulants. *Risk X: Avoid*

Tibolone: May increase anticoagulant effects of Anticoagulants. *Risk C: Monitor*

Tipranavir: May increase anticoagulant effects of Anticoagulants. *Risk C: Monitor*

Tobacco (Smoked): May decrease serum concentrations of Heparin. *Risk C: Monitor*

Urokinase: May increase anticoagulant effects of Anticoagulants. Management: Consider avoiding this combination due to an increased risk of hemorrhage. If anticoagulants are coadministered with urokinase, monitor patients closely for signs and symptoms of bleeding. *Risk D: Consider Therapy Modification*

Vitamin E (Systemic): May increase anticoagulant effects of Anticoagulants. *Risk C: Monitor*

Vitamin K Antagonists: Anticoagulants may increase anticoagulant effects of Vitamin K Antagonists. *Risk C: Monitor*

Volanesorsen: May increase anticoagulant effects of Anticoagulants. *Risk C: Monitor*

Vorapaxar: May increase adverse/toxic effects of Anticoagulants. More specifically, this combination is expected to increase the risk of bleeding. *Risk X: Avoid*

Zanubrutinib: May increase adverse/toxic effects of Anticoagulants. *Risk C: Monitor*

Pregnancy Considerations

Heparin does not cross the placenta (ESC [Regitz-Zagrosek 2018]).

Due to pregnancy-induced physiologic changes, the risk of thromboembolism is increased during pregnancy and the immediate postpartum period. Heparin may be used for anticoagulation in pregnancy (ACOG 2018). Due to a better safety profile and ease of administration, the use of low molecular weight heparin (LMWH) is generally preferred over heparin (unfractionated heparin [UFH]) in pregnancy (ACOG 2018; Bates 2018; ESC [Regitz-Zagrosek 2018]); however, heparin may be preferred in pregnant patients with kidney dysfunction (Bates 2018). Anticoagulant therapy for the prevention and treatment of thromboembolism in pregnant patients can be discontinued prior to induction of labor or a planned cesarean delivery (Bates 2018) or LMWH can be converted to UFH in higher risk patients (ESC [Regitz-Zagrosek 2018]). Consult current recommendations for appropriate use in pregnancy.

Patients with mechanical heart valves have an increased risk of adverse maternal and fetal outcomes and these risks are greater without appropriate anticoagulation. UFH or LMWH may be used in pregnant patients with mechanical heart valves. Increased monitoring is required to maintain adequate therapeutic concentrations during pregnancy (consult current recommendations for details) (ESC [Regitz-Zagrosek 2018]; ACC/AHA [Otto 2021]).

Some products contain benzyl alcohol as a preservative; their use in pregnant patients is contraindicated by some manufacturers; use of a preservative-free formulation is recommended.

Breastfeeding Considerations

Heparin is not present in breast milk (Bates 2018).

Heparin is considered acceptable for use in patients who are breastfeeding (Bates 2018; ESC [Regitz-Zagrosek 2018]). However, some products contain benzyl alcohol as a preservative; their use in breastfeeding patients is contraindicated by some manufacturers due to the association of gasping syndrome in premature infants.

Monitoring Parameters

Hemoglobin, hematocrit, platelet count, PT, aPTT, signs/symptoms of bleeding, risk factors for bleeding, fecal occult blood test (if clinically indicated); potassium.

Level of anticoagulation can be monitored by anti-Factor Xa activity **or** aPTT (calibrated by anti-Factor Xa activity or by protamine titration assay) **or** activated clotting time depending upon the indication (ACCP [Garcia 2012]; ACCP [You 2012]; Bates 2001; Hirsh 1994; Vandiver 2012).

Patients with antiphospholipid syndrome may have a prolonged aPTT at baseline due to effects of the antiphospholipid antibodies. In order to prevent a prolonged value from being mistaken for therapeutic anticoagulation, aPTT should be measured at baseline. In this situation, anti-Factor Xa monitoring may be preferred (Hull 2019a).

Platelet count should be routinely monitored to assess for risk of heparin-induced thrombocytopenia (HIT). If the patient experienced HIT within the past 100 days after receiving heparin or low-molecular-weight heparin, risk of recurrence is higher. Monitor closely if pre-exposure history is uncertain (ACCP [Guyatt 2012]).

Reference Range

Adult:

Treatment of venous thromboembolism and atrial fibrillation:

IV administration:

Anti-Factor Xa activity: 0.3 to 0.7 unit/mL.

Protamine titration: 0.2 to 0.4 unit/mL.

aPTT: Therapeutic range must be established in individual laboratories to target an aPTT prolongation that corresponds to anti-Factor Xa activity or protamine titration values (ACCP [Garcia 2012]; ACCP [You 2012]; Bates 2001; Hirsh 1994; Vandiver 2012).

SUBQ administration:

Anti-Factor Xa activity: 0.3 to 0.7 unit/mL or equivalent aPTT, obtained 6 to 8 hours after injection (ACOG 2018; Dager 2018).

ST-elevation myocardial infarction as adjunct to fibrinolysis or when no reperfusion is planned: aPTT of 1.5 to 2 times control (or an aPTT of ~50 to 70 seconds) is recommended (ACC/AHA [Rao 2025]; Lincoff 2025). **Note:** An anti-Factor Xa target has not been established for this indication.

Non-ST-elevation acute coronary syndrome: aPTT of 1.5 to 2 times control (or an aPTT of ~50 to 70 seconds) is recommended (ACC/AHA [Rao 2025]; Cutlip 2025). **Note:** An anti-Factor Xa target has not been established for this indication.

Postmechanical heart valve replacement surgery (to transition to warfarin therapy): Some experts recommend an aPTT of 1.5 to 2 times control when bleeding risk is acceptable postoperatively and heparin is initiated (ACCP [Douketis 2022]). **Note:** An anti-Factor Xa target has not been established for this indication.

Mechanism of Action

Potentiates the action of antithrombin III and thereby inactivates thrombin (as well as other coagulation factors IXa, Xa, XIa, XIIa, and plasmin) and prevents the conversion of fibrinogen to fibrin; heparin also stimulates release of lipoprotein lipase (lipoprotein lipase hydrolyzes triglycerides to glycerol and free fatty acids)

Pharmacokinetics (Adult Data Unless Noted)

Note: Increased interpatient variability of pharmacokinetic parameters in pediatric patients compared to adults; however, age-related decreases in volume of distribution and clearance with increasing pediatric patient age have been reported (ACCP [Monagle 2012]; McDonald 1981).

Onset of action: Anticoagulation: IV: Immediate; SUBQ: ~20 to 30 minutes.

Absorption: Oral, rectal: Erratic at best from these routes of administration; SUBQ absorption is also erratic, but considered acceptable for prophylactic use.

Distribution:

Premature neonates (data based on single dose of 100 units/kg within 4 hours of birth) (McDonald 1981): Inversely proportional to gestational age (GA).

GA 25 to 28 weeks: 81 ± 41 mL/kg.

GA 29 to 32 weeks: 73.3 ± 24.8 mL/kg.

GA 33 to 36 weeks: 57.8 ± 32.2 mL/kg.

Adults: Following a single 75 unit/kg dose: 36.6 ± 7.4 mL/kg (McDonald 1981).

Metabolism: Complex; thought to occur by depolymerization and desulphation via the reticuloendothelial system primarily in the liver and spleen (ACCP [Garcia 2012]; Dawes 1979; Estes 1980; Kandrotas 1992).

Half-life elimination:

Age-related: Shorter half-life reported in premature neonates compared to adult patients.

Premature neonates GA 25 to 36 weeks (data based on single dose of 100 units/kg within 4 hours of birth): Mean range: 35.5 to 41.6 minutes (McDonald 1981).

Dose-dependent: IV bolus: 25 units/kg: 30 minutes (Bjornsson 1982); 100 units/kg: 60 minutes (de Swart 1982); 400 units/kg: 150 minutes (Olsson 1963).

Mean: 1.5 hours; Range: 1 to 2 hours; affected by obesity, renal function, malignancy, presence of pulmonary embolism, and infections.

Note: At therapeutic doses, elimination occurs rapidly via nonrenal mechanisms. With very high doses, renal elimination may play more of a role; however, dosage adjustment remains unnecessary for patients with renal impairment (Kandrotas 1992).

Excretion: Urine (small amounts as unchanged drug); **Note:** At therapeutic doses, elimination occurs rapidly via nonrenal mechanisms. With very high doses, renal elimination may play more of a role; however, dosage adjustment remains unnecessary for patients with renal impairment (Kandrotas 1992).

Clearance: Age-related changes; within neonatal population, slower clearance with lower GA; however, when compared to adults, the overall clearance in neonatal and pediatric patients is faster than adults (ACCP [Monagle 2012]; McDonald 1981).

Pharmacokinetics: Additional Considerations (Adult Data Unless Noted)

Altered kidney function: The half-life may be increased.

Hepatic function impairment: The half-life may be increased or decreased.

Older adult: Plasma levels may be higher.

Patient Counseling Points

(For additional information see "Heparin (unfractionated): Patient drug information")

What is this drug used for?

- It is used to thin the blood so that clots will not form.
- It is used to treat blood clots.
- It is used to keep blood from clotting in catheters.
- It may be given to you for other reasons. Talk with the doctor.

WARNING/CAUTION: Even though it may be rare, some people may have very bad and sometimes deadly side effects when taking a drug. Tell your doctor or get medical help

right away if you have any of the following signs or symptoms that may be related to a very bad side effect:

- Bleeding like vomiting blood or vomit that looks like coffee grounds; coughing up blood; blood in the urine; black, red, or tarry stools; bleeding from the gums; abnormal vaginal bleeding; bruises without a reason or that get bigger; or any severe or persistent bleeding.
- Weakness on 1 side of the body, trouble speaking or thinking, change in balance, drooping on one side of the face, or blurred eyesight
- Swelling, warmth, numbness, change of color, or pain in a leg or arm
- Severe dizziness
- Passing out
- Confusion
- Severe headache
- Coughing up blood
- Groin or pelvic pain and swelling
- Chills
- Chest pain
- Back pain
- Shortness of breath
- Injection site skin discoloration or skin breakdown
- Signs of an allergic reaction, like rash; hives; itching; red, swollen, blistered, or peeling skin with or without fever; wheezing; tightness in the chest or throat; trouble breathing, swallowing, or talking; unusual hoarseness; or swelling of the mouth, face, lips, tongue, or throat.

Note: This is not a comprehensive list of all side effects. Talk to your doctor if you have questions.

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Brand Names: International

International Brand Names by Country

For country code abbreviations (show table)

- (AE) United Arab Emirates: Heparin leo | Hikma heparin | Liquemin | Multiparin;
- (AR) Argentina: Heparina | Heparina duncan | Heparina gemepe | Heparina sodica | Heparina veinfar | Riveparin | Riveparin-heparina rivero | Sobrius | Sodiparin | Synteparin;
- (AT) Austria: Heparin gilvasan | Heparin immuno | Ivorat anti-xa | Liquemin;
- (AU) Australia: Heparin | Heparin sodium chloride | Heparinised Saline;
- (BD) Bangladesh: Heparin | Heparon;

- (BE) Belgium: Calparine | Heparine | Heparine leo | Heparine natrium b.braun | Heparine novo-nordisk | Liqueumine;
- (BF) Burkina Faso: Heparine medis | Sakarin 25000;
- (BG) Bulgaria: Heparin | Heparin na;
- (BR) Brazil: Actparin | Cellparin | Hepamax s | Heparin | Heparina sodica | Heptar | Liqueumine | Parinex;
- (CH) Switzerland: Calciparine | Heparin | Heparin bichsel | Heparin fresenius | Heparin kabi | Heparin na | Heparin novonordisk | Heparin sintetica | Heparine choay | Liqueumin;
- (CI) Côte d'Ivoire: Pan heparine | Sakarin 25000;
- (CL) Chile: Heparina;
- (CN) China: Heparin sod.;
- (CO) Colombia: Hepamax s | Heparina | Heparina northia | Heparina sodica | Inhepar | Liqueumine;
- (CR) Costa Rica: Heparina | Heparina sodica;
- (DE) Germany: Calciparin | Calciparine | Calcium Heparin Sanofi-Aventis | Heparin | Heparin Natrium | Heparin-calcium | Liqueumin | Liqueumin n | Thrombophob | Vetren;
- (DO) Dominican Republic: Hepagel | Heparina;
- (EC) Ecuador: Hepamax s | Heparin | Heparina | Heparina sodica | Inhepar | Nuparin | Sobrius;
- (EE) Estonia: Calciparin | Heparin | Heparin sodium rovi | Heparin wzf | Heparina sodica sala | Vasparin;
- (ES) Spain: Calciparina choay | Heparina leo | Heparina sodica chiesi | Heparina sodica pan quim | Heparina sodica rovi;
- (ET) Ethiopia: Heparin;
- (FI) Finland: Hepaflex | Heparin | Heparin leo | Thromboliquine;
- (FR) France: Calciparine | Heparine Calcium | Heparine calcium dakota | Heparine calcium panpharma | Heparine roche | Heparine Sodium | Heparine Sodium Panpharma | Heparine sodium roche | Prontoprime;
- (GB) United Kingdom: Calciparine | Heparin | Heparin boots | Heparin ca | Heparin ebl | Heparin immuno | Heparin na | Heparin pain/byrne | Heparin sodium rovi | Heparin weddel | Minihep | Minihep calcium | Pump-hep | Trav heparin | Uniparin;
- (GR) Greece: Calciparine | Heparin | Heparin bichsel | Heplok;
- (HK) Hong Kong: Heparin | Heparin novo | Heparinized saline;
- (HR) Croatia: Heparin Belupo;
- (ID) Indonesia: Hepagusan | Heparin | Heparin Sodium B Braun | Heprocap | Inviclot;
- (IE) Ireland: Calciparine | Heparin | Heparin na | Heplok | Minihep | Minihep calcium | Multiparin;
- (IL) Israel: Heparin;
- (IN) India: Caprin | Cath-flush | Celhep | Celparin | Declot | Hep | Hep lock | Hepaglan | Heparen | Heparin | Hethin f | Keparin | Lofh | Lomorin | Neporin | Oxyrin h | Reniparin | Rinhepa | Thinla | Thiparin h | Thromboparin | Troyhep | V-parin;
- (IT) Italy: Calciparina | Croneparina | Ecafast | Ecasolv | Emoklar | Eparical | Eparina | Eparina calcica merck generics | Eparina calcica ratiopharm | Eparina calcica sandoz | Eparina novo | Eparina roberts | Eparina squibb | Eparina vister | Eparinger | Eparinlider | Eparven | Epsoclar | Eudipar | Flusolv | Heparinum | Liqueumin | Normoparin | Pharepa | Reoflus | Sosefluss | Trombolisin | Zepac;
- (JO) Jordan: Heparin | Hikma heparin;
- (JP) Japan: Caprocin | Deribadex | Hepacarin | Hepafilled | Hepaflush | Heparin ca sawai | Heparin calcium ajinomoto | Heparin calcium ay | Heparin lente novo | Heparin mochida | Heparin na | Heparin na lock | Heparin organon | Heparin pf novo | Heparin pfizer | Heparin sodium ajinomoto | Heparin sodium fujo | Heparin sodium fuso | Heparin sodium lock | Heparin sodium mitsubishi | Heparin sodium mochida | Heparin sodium n | Heparin sodium n ay | Heparin sodium sankyo | Novo Heparin | Novo heparin aventis | Panheprin | Pemiroc;
- (KE) Kenya: Beparine | Bruhep | Celparin | Hepa | Heparen | Heparin indar | Heparin sodium fresenius | Heparine | Heparinol | Heparovit h | Medihep | Racil;
- (KR) Korea, Republic of: Bm heparin sodium | Calciparin | Gcc heparin sodium | Hanlim heparin sodium | Heparin | Heparin na | Huons heparin sodium | Huparin | Jw heparin | Jw heparin sodium | K heparin | Pain | Pine;
- (KW) Kuwait: Heparin leo;

- (LB) Lebanon: Heparine | Heparine medis | Pharepa;
- (LT) Lithuania: Calcium heparin | Heparin Natrium | Heparin panpharma | Heparin sanofi | Heparin sodica b. braun | Heparin sodium panpharma | Heparina sodica b braun | Heparinum wzf | Nevparin;
- (LU) Luxembourg: Calparine | Heparine | Liqueumine;
- (LV) Latvia: Calcium heparin | Heparin | Heparin Natrium | Heparinum wzf | Nevparin;
- (MA) Morocco: Calciparine | Heparine sodique roche | Liqueumine;
- (MX) Mexico: Dixaparine | Hep-tec | Heparina | Heparina gi pisa | Inhepar | Proparin | Rimbipar;
- (MY) Malaysia: Eprin | Heparin | Heparinol | Unihepa | Vaxcel heparin sodium;
- (NG) Nigeria: Heparin;
- (NL) Netherlands: Calparine | Heparine;
- (NO) Norway: Hepaflex | Heparin | Heparin campus | Heparin gilvasan | Heparin leo | Heparin panpharma | Monoparin | Nycoheparin | Pump-hep;
- (NZ) New Zealand: Heparin | Heparin sodium panpharma | Monoparin | Multiparin;
- (PA) Panama: Heparina | Heparina sodica | Inhepar;
- (PE) Peru: Heparina Sodic | Heparina sodica | Paripharm | Pine | Sobrius;
- (PH) Philippines: Britton heparin na | Bruhep | Cenprahep | Dega Heparin sodium (bovine) | Hemastat | Heparin | Hepasino | Hepastal | Heprin | Kabihep | Nuparin | Sakarin 25000;
- (PK) Pakistan: Ecafast | Heparin | Pine inj. | Vaxcel heparin sodium;
- (PL) Poland: Calciparine | Coaparin | Heparin | Heparinum | Heparinum wzf;
- (PR) Puerto Rico: Hep lock | Heparin | Heparin lock flush | Heparin sodium;
- (PT) Portugal: Heparina choay | Heparina sodica b braun | Heparina sodica rovi | Heparina zentiva;
- (PY) Paraguay: Actparin | Heparina leo | Heparina sodica fada pharma | Heparina sodica fusa | Heparina sodica prosalud | Rhoneparina sodica | Sobrius;
- (QA) Qatar: Cal-Heparine | Heparin Leo | Multiparin;
- (RO) Romania: Heparin belmed;
- (RU) Russian Federation: Heparin | Heparin rus | Heparin sandoz | Heparin sodium braun;
- (SA) Saudi Arabia: Heparin | Hikma heparin | Multiparin | Vaxcel heparin sodium;
- (SE) Sweden: Heparin | Heparin Fermenta | Heparin leo;
- (SG) Singapore: Heparin;
- (SI) Slovenia: Heparin | Heparin gilvasan;
- (SK) Slovakia: Heparin;
- (SR) Suriname: Heparin | Heparine leo | Heparine natrium b.braun;
- (SV) El Salvador: Heparina sodica saimed;
- (TH) Thailand: Hepamax s | Heparin | Heparin dbi | Nuparin;
- (TN) Tunisia: Calciparine | Cutheparine | Heparine | Heparine choay | Heparine medis;
- (TR) Turkey: Biemparin | Calciparine | Heparine Sodium | Heparinum | Heparx | Hexarin | Koparin | Liqueumine | Poliparin | Seloparin | Vasparin;
- (TW) Taiwan: Agglutex | Cutheparine | Hepac | Hepac lock flush | Hepac plus | Heparin | Heparin sodium fresenius | Hesharin | Pine | Sodium heparin | T.h heparin;
- (UA) Ukraine: Gizende | Heparin;
- (UG) Uganda: Heparin;
- (UY) Uruguay: Heparina | Heparina calcica | Heparina northia | Heparina sodica | Rhoneparina sodica | Riveparin;
- (VE) Venezuela, Bolivarian Republic of: Heparina | Heparina sodica;
- (VN) Viet Nam: Henalip | Wellparin;
- (ZA) South Africa: Calciparine | Heparin;
- (ZM) Zambia: Heparin sodium fresenius;
- (ZW) Zimbabwe: Heparin | Heparin sodium fresenius

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